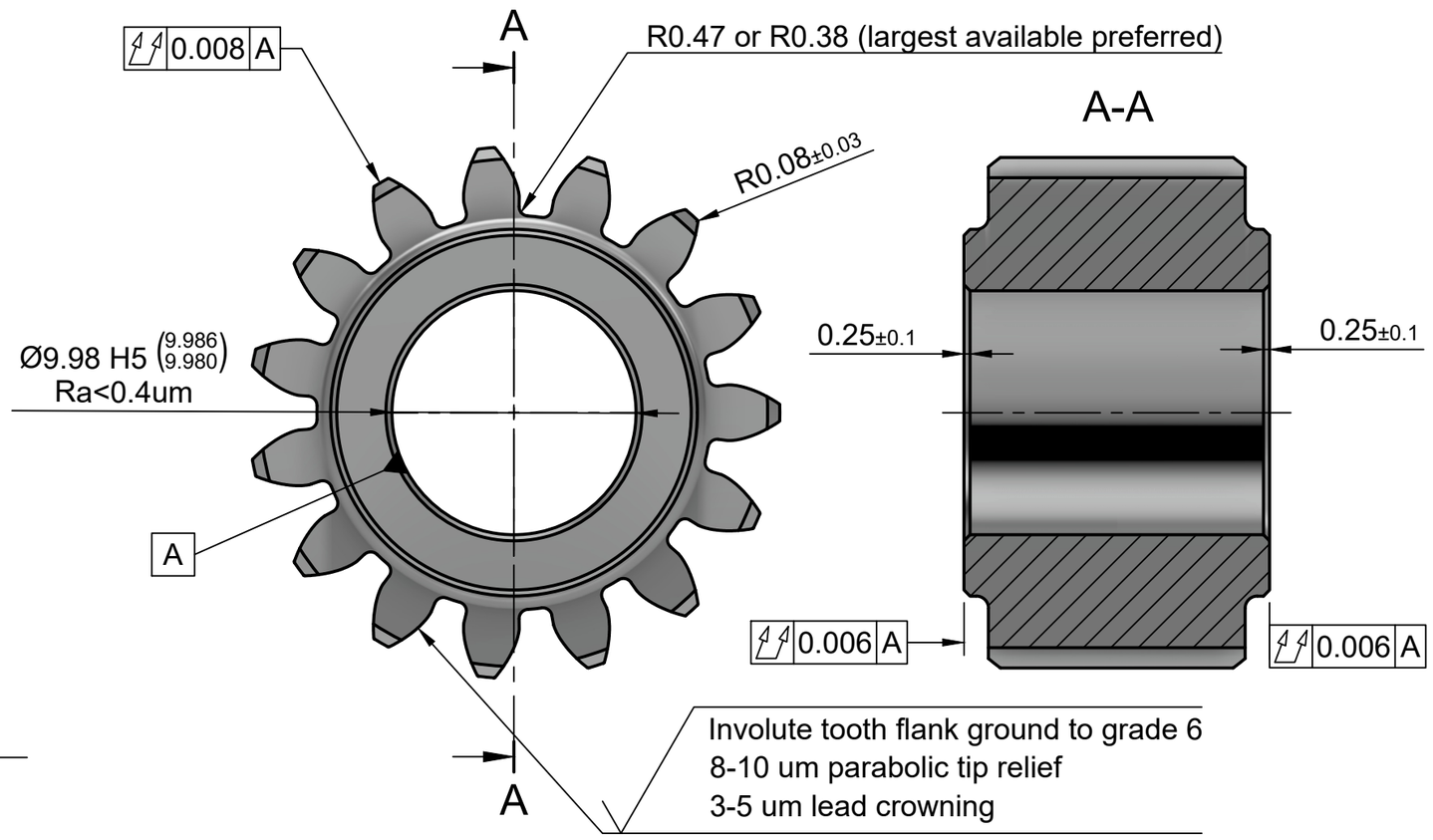
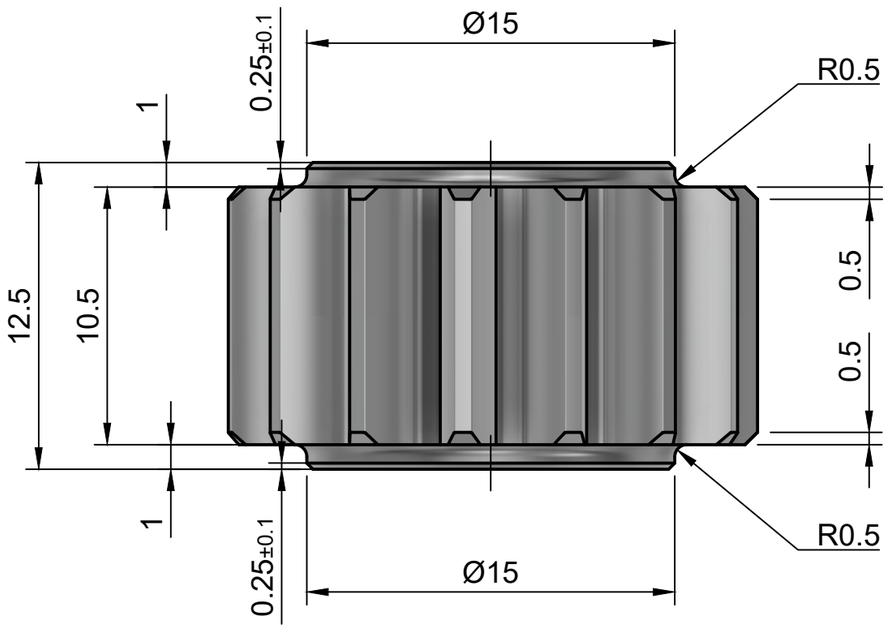




Feature	Property	Value
Module	m	1.25 mm
Number of Teeth	z	15
Pressure Angle	α	20°
Profile Shift Coefficient	x	+0.2
Reference Pitch Diameter	d	18.75 mm
Base Circle Diameter	d_b	17.619 mm
Tip Diameter	d_a	21.75 h7 mm
Root Diameter	d_f	16.125 mm
Root Fillet Radius	r_fP	0.38-0.47 mm
Tip Fillet Radius	r_tip	0.08±0.03 mm
Active Face Width	b	10.5 mm
Tooth Thickness Tolerance	DIN 3967	h24 (0/-0.02 mm)
Bore Size	d_bore	9.98 H5 mm Ra<0.4 um
Maximum Teeth Runout	bore to teeth (F_r)	0.008 mm
Maximum Axial Runout	bore to sides	0.006 mm
Normal Tooth Thickness	s_n	2.145 (0/-0.02 mm)
Chordal Tooth Thickness	s_ny	2.141 (0/-0.02 mm)
Measuring Pin Diameter	d_D	2.250 mm
Measurement Over 2 Pins	M_d	22.269 (0/-0.039 mm)

MATERIAL, PROCESSES, TREATMENTS

4140 steel quenched & tempered to 28-32 HRC
CNC hob then profile grind teeth flanks to grade 6 (Ra<0.4 um)
Plasma / Ion Nitride treatment to 58-62 HRC 0.25-0.35mm case depth
White compound layer maximum 8 um

ORDER OF OPERATIONS

1. Quench and temper stock
2.a Hob and grind
2.b Break sharp edges 0.1mm
3. Nitride treatment

Dept.	Technical reference	Created by	Approved by
R&D	ISO 2768-mK	Bence Visztra 07/11/2026	Bence Visztra 08/11/2026
Scale 4:1 Units: mm		Document type: Part Manufacturing Drawing	Document status: Released for Quote
		Title: PINION STAGE 1 40K gear drive (15T M1.25)	DWG No.: DRV-40K-101
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